

A Decentralized MILP-ADMM Framework for P2P Energy Trading in Interconnected Microgrids

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Abstract—This paper presents a privacy-preserving and fair P2P energy trading mechanism for multi-MG system. First, a MILP framework independently optimizes each MG's operation while satisfying local multi-node network constraints. Building on this, an ADMM-based decentralized scheme facilitates coordinated energy exchange while maintaining participant privacy and autonomy. A Nucleolus-based mechanism is then integrated to distribute cooperative profits fairly based on marginal contributions, ensuring coalition stability. Simulations on an IEEE 123-bus system confirm that the approach maximizes renewable energy utilization and significantly reduces operational costs. Ultimately, the framework guarantees higher economic returns for all participants compared to standalone operation, securing the coalition's long-term integrity.

Index Terms—Peer-to-peer, ADMM, Nucleolus method, MILP, transaction fees, demand response, decentralized approach.

NOMENCLATURE

Variables

P_t^{grid}	Power bought from/sold to the grid at hour t
$P_{n,t}^{line}$	Power transfer on line n at hour t
$P_{n,t}^{DG}$	Power generated by DG at node n at hour t
$C_{n,t}^{on,DG}$	The cost needed to start the DG on node n at hour t
$P_{n,t}^{chg}$	Charging power of BESS on node n at hour t
$P_{n,t}^{dis}$	Discharging power of BESS on node n at hour t
$SOC_{n,t}$	State of charge of BESS on node n at hour t
$P_{m,t}^{prop}$	Power proposal from the MG m at hour t
$P_{n,m,t}^{trade}$	Power traded from node n to MG m at hour t

Binary variables

$u_{n,t}^{chg}/u_{n,t}^{dis}$	Charging/Discharging state of BESS on node n at hour t
$u_{n,t}^{dg}$	On/Off state of DG on node n at hour t

I. INTRODUCTION

The rapid growth of Distributed Energy Resources (DERs), particularly Photovoltaic (PV) and Wind Turbine (WT) system, has driven the development of Microgrids (MGs) as a common practice for addressing modern power system operational challenges [1]. By managing local DERs and supplying loads effectively, MGs significantly enhance distribution system resilience and reduce transmission losses [2]. To optimize the utilization of DERs further, multi-MGs system increasingly adopt soft open points (SOP) to enable flexible interconnection and controllable power sharing among MGs [3]. This architectural shift enables bidirectional energy flows among interconnected MGs and facilitates Peer-to-Peer (P2P) energy trading, allowing them to exchange surplus power directly rather than relying solely on the main grid [4].

The problem of energy trading and sharing in multi-MG system has been widely and systematically studied. In particular, P2P energy trading has emerged as a key paradigm to enable decentralized, market-based coordination among autonomous MGs. In [5], Zhang et al. design a four-layer P2P architecture and auction-based pricing schemes in a community MG, showing that prosumers can reduce grid imports and energy costs while increasing local Renewable Energy Sources (RES) utilization. The study in [6] proposes a network-constrained P2P transactive energy framework for multi-MGs using a multi-leader multi-follower Stackelberg game, explicitly enforcing AC power flow limits and allowing a distributed solution for privacy preservation. More recently, [7] formulate a day-ahead Nash-bargaining-based P2P scheme for multi-MGs with data-driven chance constraints, ensuring internal network feasibility and robustness against RES and load forecast errors.

Despite the economic benefits of P2P trading, several critical barriers hinder its widespread implementation. First, MGs

are often autonomous entities owned by different stakeholders who are reluctant to share sensitive internal operational data, leading to strong privacy concerns in coordinated operation and energy sharing. Second, many existing models simplify MGs as single lumped nodes, enforcing only aggregate power balance while omitting the underlying distribution network and associated power flow, voltage and thermal constraints, which can yield schedules that violate real-world operating limits and are thus technically infeasible. Finally, traditional competitive trading mechanisms can lead to skewed benefit distributions, where exporters capture the majority of the surplus, potentially discouraging buyers and undermining the long-term stability of the trading coalition. To address these issues, this paper presents a comprehensive, privacy-preserving, and fair P2P energy trading framework for multi-MG system. The approach utilizes a Mixed-Integer Linear Programming (MILP) model to optimize each MG's local operation while strictly adhering to multi-node network constraints. A decentralized coordination layer based on the Alternating Direction Method of Multipliers (ADMM) is then employed to achieve global equilibrium through minimal information exchange, thereby preserving participant privacy. To ensure equitable participation, the Nucleolus method from cooperative game theory is integrated to redistribute cooperative profits based on each participant's marginal contribution. The contributions of this paper are three-fold as follows:

- A comprehensive MILP framework is formulated to independently optimize the operation of each individual MG. Unlike conventional studies that simplify MGs as lumped nodes, this model incorporates multi-node network constraints to ensure technical feasibility at the local level for each autonomous entity.
- An ADMM-based decentralized P2P trading scheme for a multi-MG cluster is proposed, enabling coordinated energy exchange while preserving participant privacy and autonomy.
- A fair profit-sharing scheme using the Nucleolus method is developed, which rewards each MG based on its marginal contribution and incentivizes stable, coalition-based trading.

II. METHODOLOGY

A. System Model

The system comprises m_{max} MGs ($m \in \mathcal{MG}$), each characterized by a radial tree structure with nodes $\mathcal{N} = \{0, 1, \dots, n_{max}\}$. Node 0 serves as the reference, while other nodes connect to a unique parent π_i following a topological order ($\pi_i < i$). The set of lines $\mathcal{L}$ consists of branches n connecting each node n to its parent π_i . Furthermore, each MG integrates various DERs. The locations for Distributed Generator (DG), PV, Wind, and BESS are represented by the sets $\mathcal{N}_{DG}$, $\mathcal{N}_{PV}$, $\mathcal{N}_W$, and $\mathcal{N}_B$, respectively.

B. Microgrid Self Optimization Model

The system interconnects MGs (MGs) via Soft Open Points (SOPs) to enable bidirectional energy flow, addressing the

challenges of transaction coordination and data privacy in decentralized environments. The P2P energy trading mechanism is established by integrating ADMM for distributed optimization with the Nucleolus method from cooperative game theory to ensure a stable and equitable distribution of benefits. Within this framework, the global problem is decomposed into local sub-problems for each MG. Each MG utilizes a MILP model to minimize operational costs while satisfying technical and reliability constraints. Power is supplied from the utility grid, DERs at fixed rates, and P2P trading at dynamic prices. Consequently, the local MILP model integrates internal data with signals from neighboring MGs to determine optimal schedules for DERs and Battery Energy Storage System (BESS), while simultaneously calculating the power exchange capacity for the P2P community. Through limited information exchange, the ADMM-based coordination layer iteratively updates trading prices until a global equilibrium is reached, maximizing system-wide benefits while strictly minimizing data disclosure. This problem is formulated as a MILP model which is established as follows:

1) *System power*: To ensure physical feasibility, the system is modeled using a Backward Sweep procedure (leaf-to-root) with a global power balance constraint enforced at each interval $t \in \mathcal{T}$. This ensures that the total power injection precisely matches the aggregate demand:

$$P_{n,t}^{line} + I_n^{DG} P_{n,t}^{DG} + I_n^{Wind} P_{n,t}^{Wind} + I_n^{PV} P_{n,t}^{PV} + I_n^{BESS} (P_{n,t}^{dis} - P_{n,t}^{chg}) + P_{n,m2,t}^{trade} = P_{n,t}^{Load} + \sum_{j \in \mathcal{C}_n} P_{j,t}^{line} \quad \forall n \in \mathcal{N}_{DG}, t \in \mathcal{T} \quad (1)$$

Each MG exchanges energy with the utility grid (P_t^{grid}) or neighboring MGs after meeting local demand. Grid power is decomposed into purchase (P_t^{pur}) and export (P_t^{ex}) components. To incorporate feed-in tariff (FiT) policies and capacity limits, exported power is further divided into two components: compensated power (P_t^{sell}), which can be sold to the grid, and uncompensated energy (P_t^{cur}), which must be curtailed. The compensated volume is capped at P_{max}^{ex} , representing the grid's maximum contractual or technical purchase obligation.

$$P_t^{grid} = P_{1,t}^{line} \quad (2a)$$

$$P_t^{grid} = P_t^{pur} - P_t^{ex}, \quad P_t^{pur} \geq 0, \quad P_t^{ex} \geq 0 \quad (2b)$$

$$P_t^{ex} = P_t^{sell} + P_t^{cur} \quad (2c)$$

$$0 \leq P_t^{sell} \leq P_{max}^{ex}, \quad P_t^{cur} \geq 0 \quad (2d)$$

Accordingly, the total cost associated with grid interactions is defined by (3). In this expression, λ_t^{buy} and λ_t^{sell} represent the grid-import price and the grid-export tariff, respectively.

$$J_t^{grid} = \lambda_t^{buy} P_t^{pur} - \lambda_t^{sell} P_t^{sell} \quad (3)$$

2) *DG operation constraints*: DG units enhance MG flexibility and independence through several technical constraints at node $n \in \mathcal{N}_{DG}$. These include status-dependent power limits (4a) and startup costs S_n (4b). To ensure mechanical stability, ramp-rate limits are enforced via (4c)-(4d). Further-

more, the model incorporates spinning reserve requirements (4e)-(4f) and minimum up/down time durations (4g)-(4h) to guarantee operational reliability.

$$P_n^{DG} u_{n,t}^{DG} \leq P_{n,t}^{DG} \leq \bar{P}_n^{DG} u_{n,t}^{DG}, \quad \forall t \in \mathcal{T} \quad (4a)$$

$$C_{n,t}^{on,DG} = \begin{cases} S_n \cdot (u_{n,t}^{DG} - u_{n,t-1}^{DG}) & \text{if } t > 0 \\ S_n \cdot u_{n,t}^{DG} & \text{if } t = 0 \end{cases}, \quad \forall t \in \mathcal{T} \quad (4b)$$

$$P_{n,t}^{DG} - P_{n,t-1}^{DG} \leq \Delta P_{n,DG}^{\max}, \quad \forall t \in \mathcal{T} \setminus \{0\} \quad (4c)$$

$$P_{n,t-1}^{DG} - P_{n,t}^{DG} \leq \Delta P_{n,DG}^{\max}, \quad \forall t \in \mathcal{T} \setminus \{0\} \quad (4d)$$

$$SR_{n,t}^{DG} = \bar{P}_n^{DG} u_{n,t}^{DG} - P_{n,t}^{DG}, \quad \forall t \in \mathcal{T} \quad (4e)$$

$$\sum_{n \in \mathcal{N}_{DG}} SR_{n,t}^{DG} \geq 0.2 \sum_{n \in \mathcal{N}_{DG}} \bar{P}_n^{DG}, \quad \forall t \in \mathcal{T} \quad (4f)$$

$$\sum_{\phi=t}^{t+T_{n,up}^{DG}-1} u_{n,\phi}^{DG} \geq T_{n,up}^{DG} (u_{n,t}^{DG} - u_{n,t-1}^{DG}), \quad \forall t \in \mathcal{T} \quad (4g)$$

$$\sum_{\phi=t}^{t+T_{n,down}^{DG}-1} (1 - u_{n,\phi}^{DG}) \geq T_{n,down}^{DG} (u_{n,t-1}^{DG} - u_{n,t}^{DG}), \quad \forall t \in \mathcal{T} \quad (4h)$$

The total DG operating cost comprises three primary components: fuel costs modeled as a quadratic function of $P_{n,t}^{DG}$, startup costs incurred during state transitions, and variable O&M costs proportional to the electrical output:

$$J_t^{DG, \text{total}} = \sum_{n \in \mathcal{N}_{DG}} \left(\alpha_n u_{n,t}^{DG} + \beta_n P_{n,t}^{DG} + \theta_n (P_{n,t}^{DG})^2 + C_{n,t}^{on,DG} + \kappa_n P_{n,t}^{DG} \right) \quad (5)$$

3) *BESS operation constraints*: To enable energy-time shifting and maintain supply-demand balance, BESS units $n \in \mathcal{N}_{BESS}$ operate within technical power and state-of-charge SOC limits as follows:

$$0 \leq P_{n,t}^{\text{chg}} \leq P_{n,\max}^{\text{BESS}} u_{n,t}^{\text{chg}}, \quad \forall t \in \mathcal{T} \quad (6a)$$

$$0 \leq P_{n,t}^{\text{dis}} \leq P_{n,\max}^{\text{BESS}} u_{n,t}^{\text{dis}}, \quad \forall t \in \mathcal{T} \quad (6b)$$

$$\underline{SOC}_n \leq SOC_{n,t} \leq \overline{SOC}_n, \quad \forall t \in \mathcal{T} \quad (6c)$$

$$SOC_{n,t} = SOC_{n,t-1} + \frac{1}{E_n^{\text{nom}}} \left(P_{n,t}^{\text{chg}} \eta_n - \frac{P_{n,t}^{\text{dis}}}{\eta_n} \right), \quad \forall t \in \mathcal{T} \setminus \{0\} \quad (6d)$$

$$SOC_{n,0} = SOC_{n,\text{ini}}, SOC_{n,T} = SOC_{n,\text{ini}} \quad (6e)$$

Equations (6a)-(6b) define charging and discharging power limits to prevent equipment degradation. SOC boundaries (6c) maintain safety margins, while SOC dynamics (6d) capture energy balance considering efficiency η_n . To ensure sustainable daily cycles, SOC periodicity is enforced via (6e).

Unlike conventional generators, BESS operational expenses

focus on degradation rather than fuel. A degradation coefficient θ_n is applied to the energy throughput, ensuring battery cycling only occurs when economic gains outweigh cell aging costs:

$$J_t^{\text{BESS}} = \sum_{n \in \mathcal{N}_B} \left(\theta_n \times (\eta_n P_{n,t}^{\text{chg}} + \frac{P_{n,t}^{\text{dis}}}{\eta_n}) \right) \quad (7)$$

4) *Trading constraints*: To facilitate energy trading between feeders, the SOP model is integrated between MGs $m1$ and $m2$. In the local optimization stage, trading power serves as a decision variable to minimize costs, subject to trading boundaries (8d) and SOP physical limits (8d). Transmission losses are modeled as a linear function using the coefficient K_m^{SOP} . Notably, the global power exchange balance is resolved through a decentralized consensus process rather than being directly computed within the local MILP sub-problem.

$$P_{m1,m2,t}^{\text{trade}} + P_{m1,m2,t}^{\text{tradeLoss}} = P_{m2,t}^{\text{prop}} + P_{m2,t}^{\text{propLoss}} \quad (8a)$$

$$\underline{P}_{tr} \leq P_{m1,m2,t}^{\text{trade}} \leq \bar{P}_{tr} \quad (8b)$$

$$P_{m1,m2,t}^{\text{Loss}} = K_m^{\text{SOP}} \cdot P_{m1,m2,t}^{\text{trade}} \quad (8c)$$

$$|P_{m1,m2,t}^{\text{trade}}| \leq P_{m1,m2}^{\text{SOP}} \quad (8d)$$

The trading cost is formulated as the product of the energy exchange volume and the price signal $\lambda_{m1,m2,t}^{\text{trade}}$ received from the centralized coordinator. This price is subjected to the boundaries defined in (9a), with the overall cost calculation established in (9b).

$$\lambda_t^{\text{sell}} \leq \lambda_{m1,m2,t}^{\text{trade}} \leq \lambda_t^{\text{buy}} \quad (9a)$$

$$J_t^{\text{trade}} = \sum_{m \in \mathcal{M}} (\lambda_{m1,m2,t}^{\text{trade}} \cdot P_{m1,m2,t}^{\text{trade}}) \quad (9b)$$

5) *Objective function*: The objective of this local sub-problem is to minimize the individual operational cost for each MG through a self-optimization process. As formulated in equation (10), the total cost function aggregates the expenses from utility grid interactions, DG operation, BESS degradation, and P2P trading transactions.

$$\min \sum_{t \in \mathcal{T}} \left(J_t^{\text{grid}} + J_t^{\text{DG}} + J_t^{\text{BESS}} + J_t^{\text{trade}} \right) \quad (10)$$

s.t. Constraints (1), (4), (6), (8b)-(8d)

C. Decentralized P2P Energy trading between Multi-MGs

The global objective extends (10) for all $m \in \mathcal{M}$ subject to power trading equilibrium. Assuming negligible SOP losses, the trading balance simplifies to (11b), while prices are bounded to ensure market stability.

$$\min \sum_{m \in \mathcal{M}} \sum_{t \in \mathcal{T}} \left(J_t^{\text{grid}} + J_t^{\text{DG}} + J_t^{\text{BESS}} + J_t^{\text{trade}} \right) \quad (11a)$$

s.t. Constraints (1), (4), (6), (8), (9a)

$$P_{m1,m2,t}^{\text{trade}} = P_{m2,t}^{\text{prop}} \quad (11b)$$

The coordinator maps local data to global auxiliary variables

$\hat{p}_{m2,t}^{m1}$ to ensure trading balance.

$$\hat{p}_{m2,t}^{m1} = P_{m1,m2,t}^{trade}, \quad \hat{p}_{m2,t}^{m1} + \hat{p}_{m1,t}^{m2} = 0 \quad (12)$$

Defining x_t^m as the local proposal vector and y_t^m as the auxiliary set, the objective function at equation (10) is augmented with Lagrangian multipliers λ_t^m to penalize consensus violation:

$$\begin{aligned} \mathcal{F}_m(x_t^m, y_t^m, \lambda_t^m) = & \sum_{t \in \mathcal{T}} \left\{ \left(J_{m,t}^{grid} + J_{m,t}^{DG} + J_{m,t}^{BESS} + J_{m,t}^{trade} \right) \right. \\ & \left. + \sum_{m1,m2 \in \mathcal{M}} \left(\lambda_{m2,t}^{m1} \cdot \Delta P_{m2,t}^{m1} + \frac{\rho}{2} \|\Delta P_{m2,t}^{m1}\|_2^2 \right) \right\} \\ \text{s.t. Constraints (1), (4), (6), (8b)-(8d)} \\ \text{with } \Delta P_{m2,t}^{m1} = & P_{m1,m2,t}^{trade} - \hat{p}_{m2,t}^{m1} \end{aligned} \quad (13)$$

Let denote i is the iteration index. Then the procedure based ADMM method as follow:

1) *Update x_t^m* : $x_t^{m,i+1}$ can be update locally by 2 fixing the auxiliary variables $y_t^{m,i}$ and Lagrangian multiplier $\lambda_t^{m,i}$:

$$x_t^{m,i+1} = \underset{x_t^m}{\operatorname{argmin}} \mathcal{F}_m(x_t^m, y_t^m, \lambda_t^m) \quad (14)$$

2) *Update y_t^m* : Upon receive x_t^m and λ_t^m , MGs $m1$ and $m2$ update y_t^m or $\hat{p}_{m2,t}^{m1}$ as follow:

$$\begin{aligned} \hat{p}_{m2,t}^{m1,i+1} = -\hat{p}_{m1,t}^{m2,i+1} = & \frac{P_{m1,m2,t}^{trade} - P_{m2,m1,t}^{trade}}{2} \\ & + \frac{\lambda_{m1,t}^{m2} - \lambda_{m2,t}^{m1}}{2\rho} \end{aligned} \quad (15)$$

3) *Update λ_t^m* : The Lagrangian multipliers has update local by using $x_t^{m,i+1}$ and $y_t^{m,i+1}$ and then sending the data of $y_t^{m,i+1}$ and $\lambda_t^{m,i+1}$ to trading MG.

$$\lambda_{m2,t}^{m1,i+1} = \lambda_{m2,t}^{m1,i} + \rho \cdot \Delta P_{m2,t}^{m1} \quad (16)$$

4) *Update ρ* : To mitigate sensitivity to initial parameter tuning and guarantee algorithmic stability, ρ is dynamically adjusted based on the primal-dual residual ratio, ensuring rapid convergence.

D. Cooperative Game Theory for Profit Sharing: The Nucleolus Approach

To ensure equitable benefit distribution, this study employs the Nucleolus concept, solved via sequential Linear Programming to minimize maximum coalition dissatisfaction. The problem is modeled as a cooperative game $(\mathcal{MG}, v)$, where $\mathcal{MG}$ denotes the set of players and $g \in \mathcal{G}$ represents the profit allocation. For any coalition $S \subseteq \mathcal{MG}$, the characteristic value $v(S)$ represents the maximum collective profit from internal trading, determined by executing the ADMM algorithm for subset S .

The Nucleolus is computed via sequential LPs. Stage 1 (P_1) minimizes the maximum excess ϵ_1 subject to efficiency and

individual rationality constraints:

$$\min \epsilon_1 \quad \text{s.t.} \quad \begin{cases} \sum_{m \in \mathcal{MG}} x_m = v(\mathcal{MG}); & x_m \geq v(\{m\}) \\ v(S) - \sum_{m \in S} x_m \leq \epsilon_1, & \forall S \subset \mathcal{MG} \end{cases} \quad (17)$$

If the solution is non-unique, subsequent stages k minimize the next largest excess ϵ_k for remaining coalitions $\mathcal{U}_{k-1}$, while fixing settled excesses for $\mathcal{F}_j$ from previous steps:

$$\min \epsilon_k \quad \text{s.t.} \quad \begin{cases} v(S) - \sum_{m \in S} x_m = \epsilon_j^*, & \forall S \in \mathcal{F}_j, j < k \\ v(S) - \sum_{m \in S} x_m \leq \epsilon_k, & \forall S \in \mathcal{U}_{k-1} \end{cases} \quad (18)$$

III. NUMERICAL RESULTS

The proposed P2P trading algorithm is evaluated using the IEEE 123-bus distribution network (DN), as illustrated in Fig. 1, which consists of four interconnected MGs. The detailed technical parameters for the DG and BESS are summarized in Table I, while the Time-of-Use (TOU) grid pricing mechanism is presented in Fig. 3.

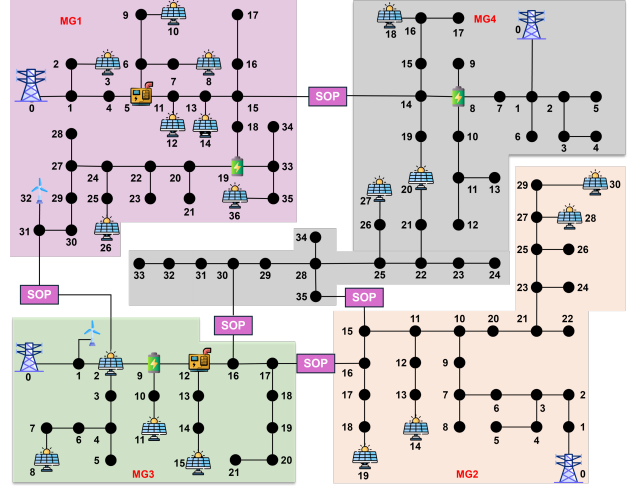


Fig. 1: Layout of the four interconnected MGs system

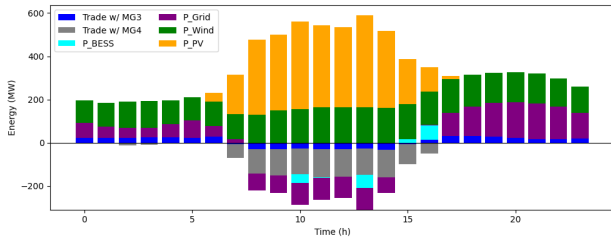
Assuming 100% RES utilization with 60 kW PV and 170 kW Wind capacities, the proposed algorithm proves highly efficient: evaluating all 15 coalitions takes only 149.6s, while the ADMM with dynamic ρ update converging within 4-21 iterations (≤ 33.63 s/subset). Consequently, the simulation results are comprehensively analyzed based on the Validation of Operational Feasibility and Modeling, Economic Performance Analysis, and Economic Fairness and Individual Rationality.

A. Validation of Operational Feasibility and Modeling

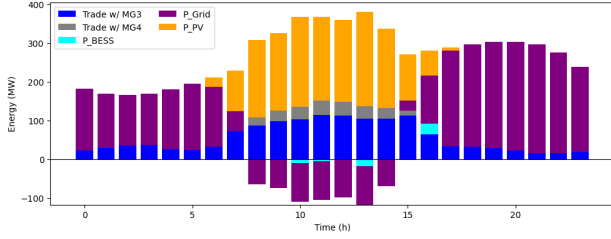
The results of the 24-hour optimal energy plan for four interconnected MGs are represented in Fig. 2. Instead of simplifying everything, this simulation clearly shows the self balancing energy and exact trades between different MGs. The

TABLE I: DG and BESS

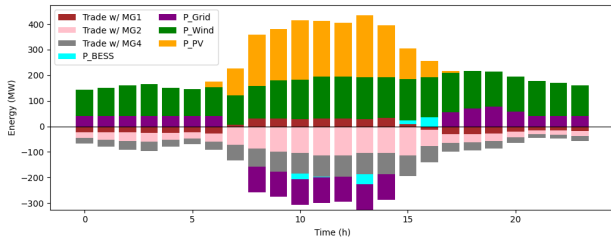
DG		BESS	
Parameter	Value	Parameter	Value
P^{DG}	20	(E_n^{nom})	150
$\bar{P}^{DG}$	130	η	0.9
$\Delta P_{DG}^{\text{max}}$	100	$P_{n,\text{max}}^{\text{BESS}}$	75
$T_{up}^{DG}/T_{down}^{DG}$	2	$SOC_{\min}$	0.2
C_{DG}^{hot}	660	$SOC_{\max}$	0.95
C_{DG}^{cold}	1120	SOC_{ini}	0.4
α	700	θ_{BESS}	0.2
β	16.06		
θ	0.002		
κ	1.258		



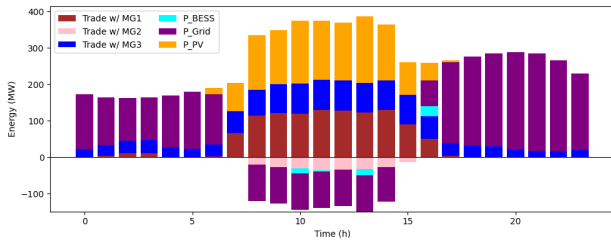
(a) Microgrid 1



(b) Microgrid 2



(c) Microgrid 3



(d) Microgrid 4

Fig. 2: Energy trading results for four interconnected MGs

generation heterogeneity across the network is fully exploited MG3 has a lot of wind power but not much load, so it acts as the main power source, which exports excess energy to other MGs. Conversely, with limited renewable capacity, rely on these P2P imports to meet demand, MG1 plays a dual role, acting as an exporter during peak irradiance (08:00 - 15:00) and an importer during off-peak hours. This strategy aligns with the SOC behavior in Fig. 3 From 09:00 to 12:00, MGs store solar energy to avoid low-cost exports and cover the retain RES power.

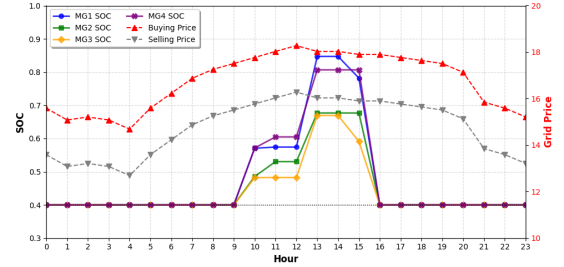


Fig. 3: SoC of BESS in four MGs

Notably, MG3 keeps a lower SOC (≈ 0.68) because it prioritizes exporting power to support the network. After 15:00, all batteries discharge to cover the evening peak, avoiding expensive grid electricity. This proves the model's ability to coordinate P2P, BESS, and grid interaction effectively.

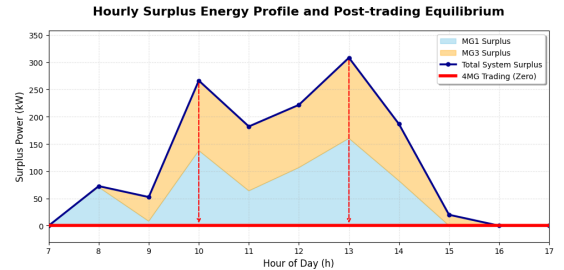


Fig. 4: RES retained independence case and trading case

Figure 4 shows that the model achieves 100% RES utilization. Without P2P, the surplus from MG1 and MG3 (peaking over 300 kW) would be wasted or sold to the grid at low prices. The proposed mechanism turns this surplus into a trading resource, where it is fully absorbed by MG2 and MG4. This eliminates RES curtailment, ensuring all green energy is consumed within the local network.

B. Economic Performance Analysis

As detailed in Table II, the P2P trading mechanism yields a substantial aggregate profit of 36,919.67 cents, with every MG achieving a reduction in operational costs. However, a closer inspection reveals a skewed benefit distribution: the vast majority of the surplus (90%) is captured by net exporters (MG1, MG3), leaving net importers (MG2, MG4) with minimal gains. This disparity underscores the limitations of a purely competitive trading scheme and motivates the

TABLE II: Total energy procurement cost/profit of each MG with/without trading (cents)

MG	Base Line	P2P Trading	P2P Trading profit
MG1	30,197.34	15,124.3	15,073.04
MG2	77,808.47	76,526.1	1,282.37
MG3	-13,994.11	-32,704.46	18,710.35
MG4	76,169.78	74,315.87	1,853.91
Total	170,181.47	133,261.8	36,919.57

subsequent application of the Nucleolus method to redistribute the cooperative gains more equitably.

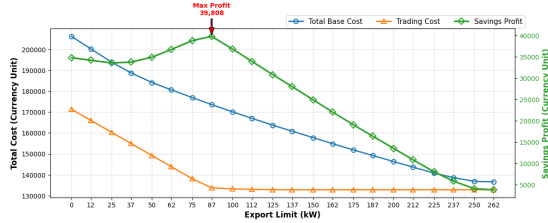


Fig. 5: The profit system based on export limit (cents)

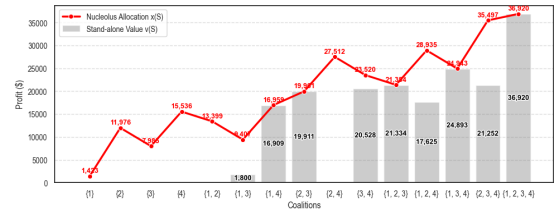
The proposed algorithm demonstrates superior performance not only at a single operating point but also through its robustness against grid policy fluctuations. Fig. 5 depicts the system's profit behavior under varying export limits (0 to >250 kW). Notably, the trading cost remains consistently lower than the base cost. Since this substantial profit margin easily absorbs practical SOP losses (2-3%), the algorithm guarantees positive savings regardless of grid strictness, even under zero-export conditions. Such dynamic adaptability validates the model's practical feasibility across various regulatory frameworks.

C. Economic Fairness and Individual Rationality

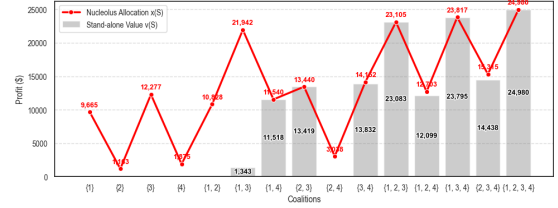
Fig. 6 demonstrates that the Nucleolus mechanism dynamically captures market power shifts governed by grid interactions. It adjusts profit distribution by prioritizing buyers during tight export constraints, while incentivizing sellers when grid access is relaxed. Unlike proportional allocation rules that ignore bargaining leverage, or the Shapley value that merely averages marginal contributions and occasionally violates Core boundaries, the Nucleolus strictly minimizes maximum coalition dissatisfaction. This guarantees the stability condition in equation (17), ensuring every participant consistently achieves higher returns than in standalone operation, thereby securing the long-term integrity of the P2P coalition.

IV. CONCLUSION

In this study, a privacy-preserving and fair P2P energy trading mechanism for multi-MG systems is developed. By decomposing the global optimization problem via ADMM, the framework allows MGs to coordinate optimal schedules without revealing sensitive operational data. The numerical simulations confirm the economic viability of the proposed



(a) Export limit: 100



(b) Export limit: 150

Fig. 6: Profit allocation using the nucleolus method

strategy, yielding a substantial total profit of 36,919 cents while effectively mitigating RES curtailment.

A key contribution of this work is the rigorous analysis of market robustness and fairness. The study demonstrates that the system maintains economic efficiency across regulatory constraints, identifying a saturation point in internal trading at an 87 kW export limit. Moreover, integrating the Nucleolus method dynamically adjusts profit distribution based on bargaining power, correcting the skew towards net exporters observed in pure markets. By exchanging only boundary variables, this decentralized approach inherently minimizes communication overhead and cyber-security risks. While the deterministic model establishes a fair coordination framework, future work will integrate stochastic programming to address renewable and load uncertainties, enhancing real-time reliability.

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